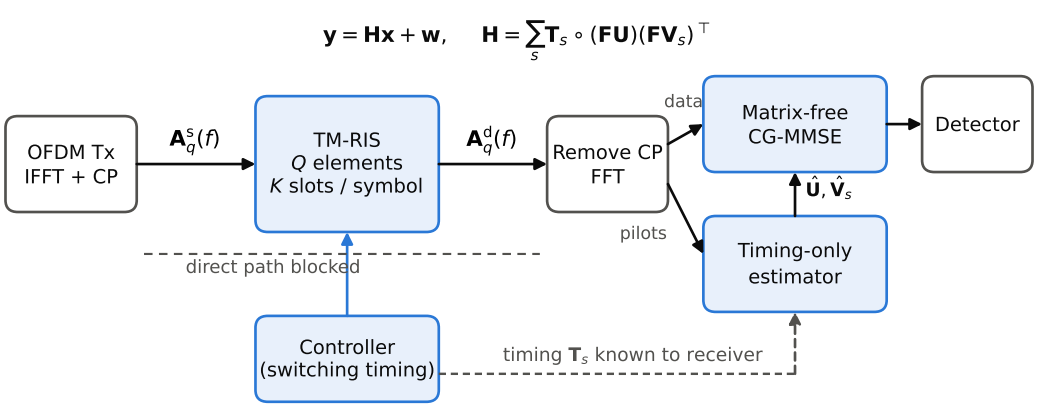
$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}, \quad \mathbf{H} = \sum_s \mathbf{T}_s \circ (\mathbf{F}\mathbf{U})(\mathbf{F}\mathbf{V}_s)^\top$
OFDM Tx
IFFT + CP
$\mathbf{A}_q^s(f)$
TM-RIS
Q elements
K slots / symbol
$\mathbf{A}_q^d(f)$
Remove CP
FFT
data
pilots
Matrix-free
CG-MMSE
Detector
$\hat{\mathbf{U}}, \hat{\mathbf{V}}_s$
Timing-only
estimator
direct path blocked
Controller
(switching timing)
timing $\mathbf{T}_s$ known to receiver